Endsem Assignment

Name: Mohd Darish Khan

Roll: 2101MC29

Q1:

Code:

```
% part 2
func = @(x) (x(1) + 10*x(2))^2 + 5*(x(3)-x(4))^2 + (x(2) -2*x(3))^4 + 10*(x(1)-
x(4))^4;
x0 = [3; -1; 0; 1];
eps = 1e-6;

[x_opt, k] = Newton_method(func, x0, eps);

disp('Minimum found at x = ');
disp(x_opt);
fprintf('Number of iterations performed = %d\n', k);

% general function
function [x_opt, k] = Newton_method(f, x0, eps)

    syms x [length(x0) 1];
    f1 = f(x);

    grad_sym = gradient(f1, x);
    hess_sym = hessian(f1, x);

    grad_func = matlabFunction(grad_sym, 'Vars', {x});
    hess_func = matlabFunction(hess_sym, 'Vars', {x});

    iter = 0;
    x_curr = x0;

    grad = grad_func(x_curr);

    while norm(grad, 2) >= eps
        grad = grad_func(x_curr);
        hess = hess_func(x_curr);

        x_curr = x_curr - hess \ grad;
        iter = iter + 1;
    end

    x_opt = x_curr;
    fprintf('function value f(x) = %f\n', f(x_opt));
    k = iter;
end
```

Output:

```
>> q1
function value f(x) = 0.000000
Minimum found at x =
    0.0016
   -0.0002
    0.0003
    0.0003

Number of iterations performed = 18
```

Q2:

Code:

```
f = @(x) (x(1)-4)^4 + (x(2)-3)^2 + 4*(x(3)+5)^4;

grad_f = @(x) [
    4 * (x(1) - 4)^3;
    2 * (x(2) - 3);
    16 * (x(3) + 5)^3
];

x0 = [0; 0.001; 0];
tol = 1e-4;
max_iter = 1000;
alpha0 = 1;

[x, iter] = steepest_descent_with_secant(f, grad_f, x0, tol, max_iter, alpha0);
fprintf('Value of x = [%s].\n', sprintf('%.6f ', x));
fprintf('number of iterations reached = %d \n', iter)
fprintf("final value of function f(x) = %f \n", f(x))


% code for secant method
function [x, iter] = steepest_descent_with_secant(f, grad_f, x0, tol, max_iter,
alpha0)
    x = x0;
    iter = 0;

    while norm(grad_f(x)) > tol && iter < max_iter
        d = -grad_f(x);

        alpha = secant_line_search(f, grad_f, 1e-6, x, alpha0, d);

        x = x + alpha * d;

        iter = iter + 1;
    end

    if norm(grad_f(x)) > tol
        fprintf('Maximum iterations reached. Approximate solution: x = [%s].\n',
sprintf('%.6f ', x));
    end
end

%code for secant line search
function alpha = secant_line_search(f, grad_f, eps, x_current, a0, d)
    a1 = a0;
    a2 = a0 * 1.1;

    phi_p = @(alpha) grad_f(x_current + alpha * d)' * d;

    while abs(phi_p(a2)) > eps
        a_new = a2 - phi_p(a2) * (a2 - a1) / (phi_p(a2) - phi_p(a1));

        a1 = a2;
        a2 = a_new;
    end
```

```
    alpha = a2;
end
```

Output:

```
>> q2
Value of x = [4.027556 2.999996 -5.013782 ].
number of iterations reached = 148
final value of function f(x) = 0.000001
```